

Surrogate-Assisted Genetic Optimization of a PSA CO₂ Capture Process

SEEN5320 Final Project

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Abstract

This project builds a closed optimization workflow for a pressure swing adsorption (PSA) CO₂ capture process using only the existing scalar simulation records in `data/manifest.csv`. Deleted time-space CSV files are not used. A manifest-only surrogate maps seven PSA operating variables to purity, recovery, productivity, and specific energy. The surrogate is then coupled with a Python implementation of NSGA-II to solve two bi-objective problems: purity versus recovery, and productivity versus energy. The optimized inputs and surrogate-predicted outputs are exported for detailed-model reruns through CSV files, completing the data handoff needed for closed-loop comparison.

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1 Objective and Workflow

The objective is to replace deleted-file-dependent analysis with a practical surrogate-assisted optimization workflow. The research route is:

1. Train surrogate models from the 1000 successful rows in `manifest.csv`.
2. Use the surrogate inside a multi-objective genetic algorithm.
3. Export optimized decision variables to the detailed PSA model.
4. Compare detailed-model outputs against surrogate predictions after the detailed reruns are available.

The workflow is intentionally manifest-only. Auxiliary file-path columns in the manifest are ignored, and only scalar design variables and KPI outputs are read.

2 Data

The manifest contains 1000 successful Zeolite 13X PSA simulations. The seven decision variables are listed in Table 1; their bounds are read from `ProcessConfig.yaml`.

Table 1: Design variables used by the surrogate and GA.

Variable	Meaning	Bound
x_1	Adsorption pressure	100000–1000000 Pa
x_2	Adsorption time	10–5000 s
x_3	Light-product reflux ratio	0–1
x_4	Feed velocity	0.1–1.0 m/s
x_5	Blowdown pressure	10000–100000 Pa
x_6	Pressurization time	0–100 s
x_7	Counter-current depressurization time	0–100 s

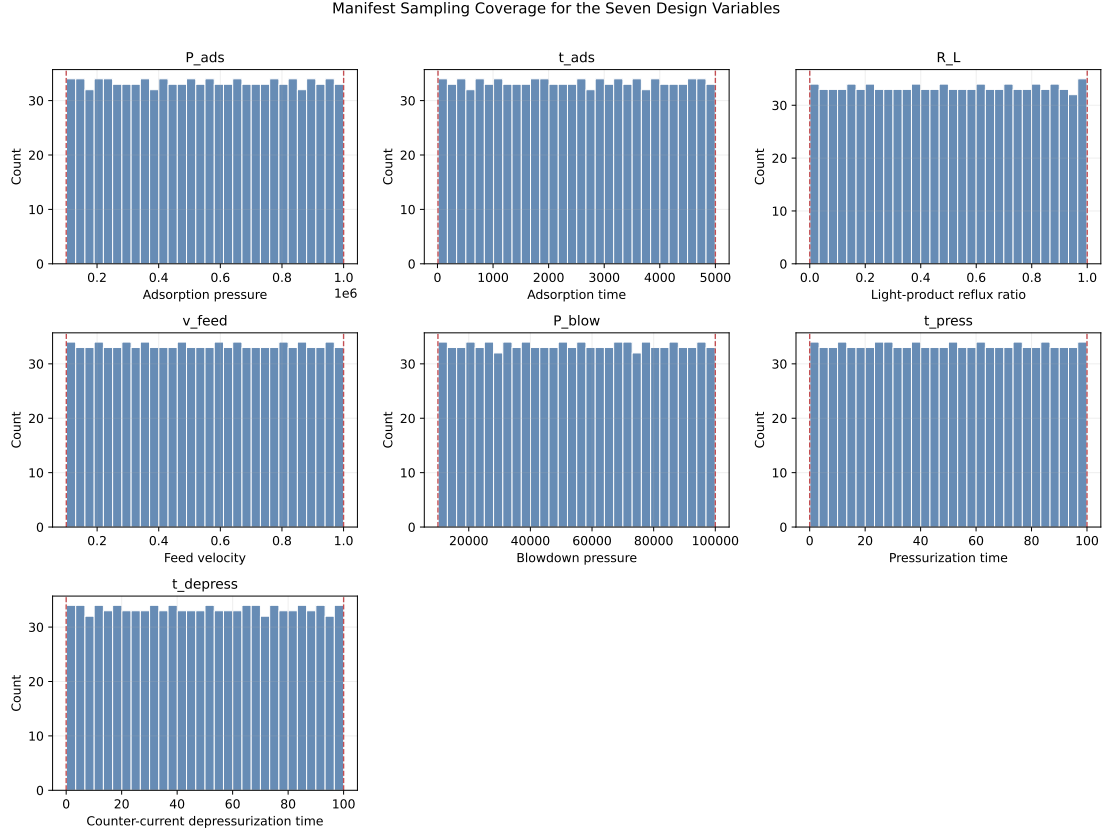


Figure 1: Manifest sampling coverage of the seven design variables.

The manifest output ranges are broad: purity spans 0.150–0.670, recovery spans 0.0056–1.000, productivity spans 0.243–115.45 mol kg^{−1} h^{−1}, and energy spans 36.70–84378.80 kWh ton^{−1}.

3 Surrogate Model

The surrogate uses a degree-3 polynomial response surface in standardized design variables. A fixed 80/20 train-test split with random seed 42 is used. Ridge regularization is selected by five-fold cross-validation on the training set.

The regression targets are CO₂ recovery, productivity, and log₁₀(specific energy). Purity is modeled as log₁₀($p - 0.15$), where p is CO₂ purity and 0.15 is the feed CO₂ fraction. This transformation improves the fit near feed-level purity. During optimization, predictions are clipped to the observed manifest output envelope to avoid reporting extrapolated KPI values outside the training data range.

Table 2: Surrogate test-set accuracy. Energy is also evaluated in log space because the physical energy values are highly skewed.

Target	RMSE	MAE	R^2
Purity	0.01658	0.00555	0.800
Recovery	0.05016	0.02329	0.973
Productivity	5.495	1.337	0.933
Energy	1621.6	381.6	0.775
$\log_{10}(\text{energy})$	0.08166	0.05896	0.964

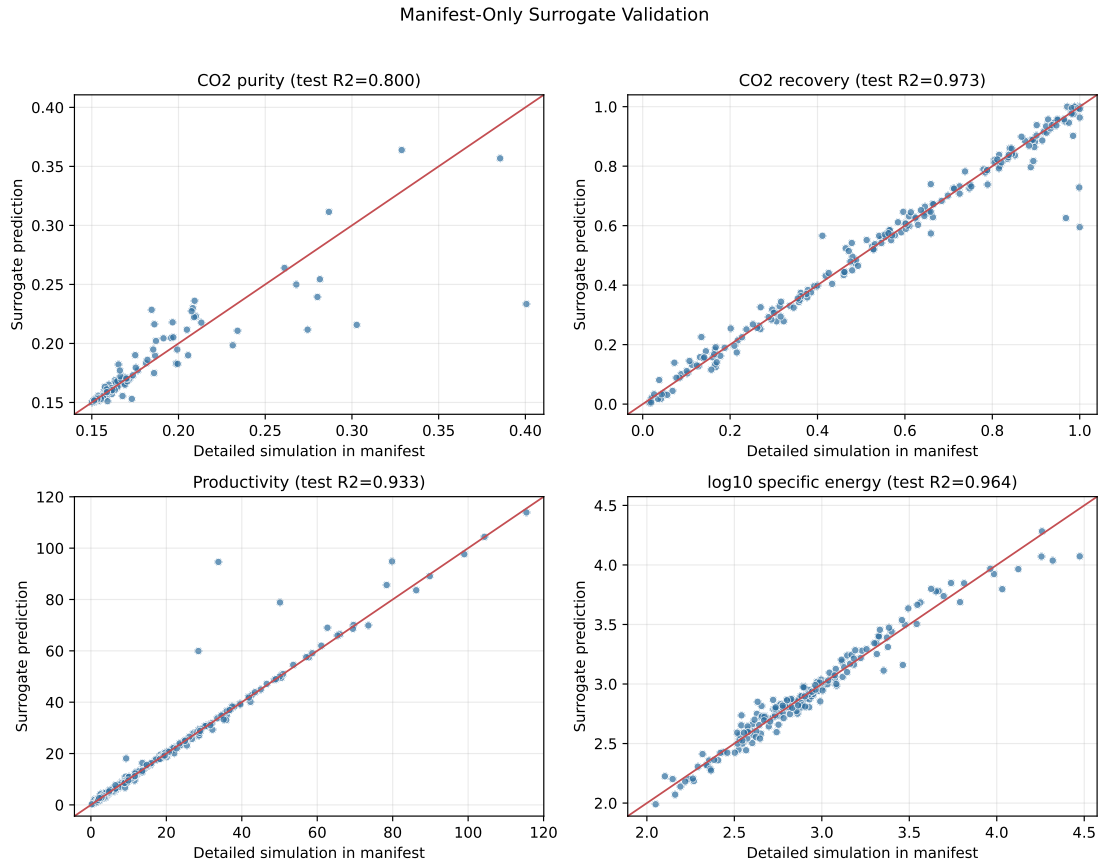


Figure 2: Predicted versus simulated values on the held-out test set.

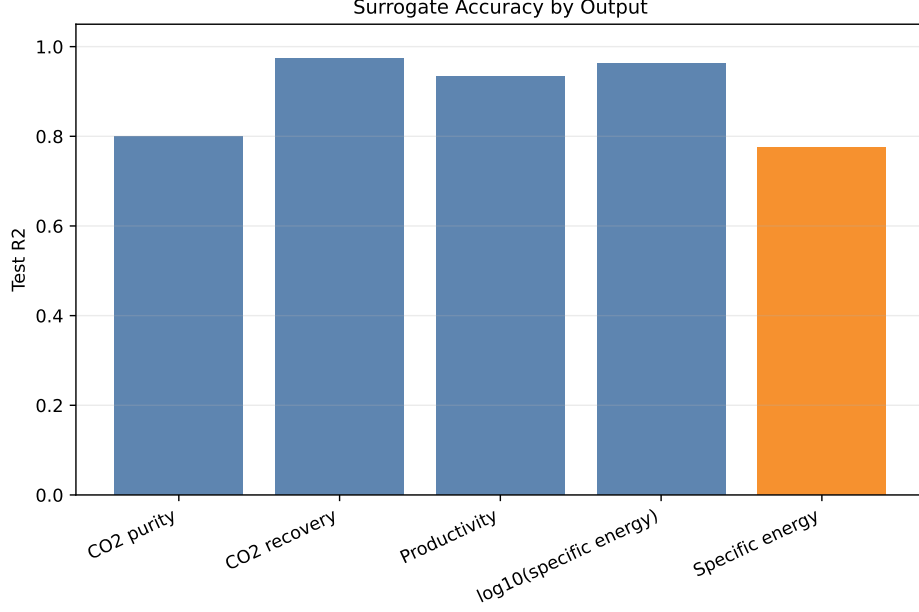


Figure 3: Test-set R^2 summary for the manifest-only surrogate.

4 Genetic Algorithm Optimization

The surrogate is optimized with an in-house NumPy implementation of NSGA-II. The GA operates directly on the seven bounded design variables and uses the surrogate for fast KPI evaluation. The settings are:

- Population size: 120.
- Generations: 200.
- Simulated binary crossover probability: 0.9.
- Polynomial mutation probability: 1/7 per variable.
- Initial population: half manifest-seeded points and half random points inside the configured bounds.

Two bi-objective optimization problems are solved:

$$\max_x [purity(x), recovery(x)],$$

and

$$\max_x [productivity(x), -\log_{10}(energy(x))].$$

The final nondominated set from each problem contains 120 surrogate Pareto candidates. Twenty representative candidates from each front are exported to `data/detailed_model_input.csv`.

5 Optimization Results

For the purity-recovery problem, the surrogate Pareto set covers predicted purity from 0.436 to 0.670 and recovery from 0.913 to 1.000. The corresponding productivity remains low, around 5.5–5.9 mol kg⁻¹ h⁻¹, showing the expected tradeoff between separation quality and throughput.

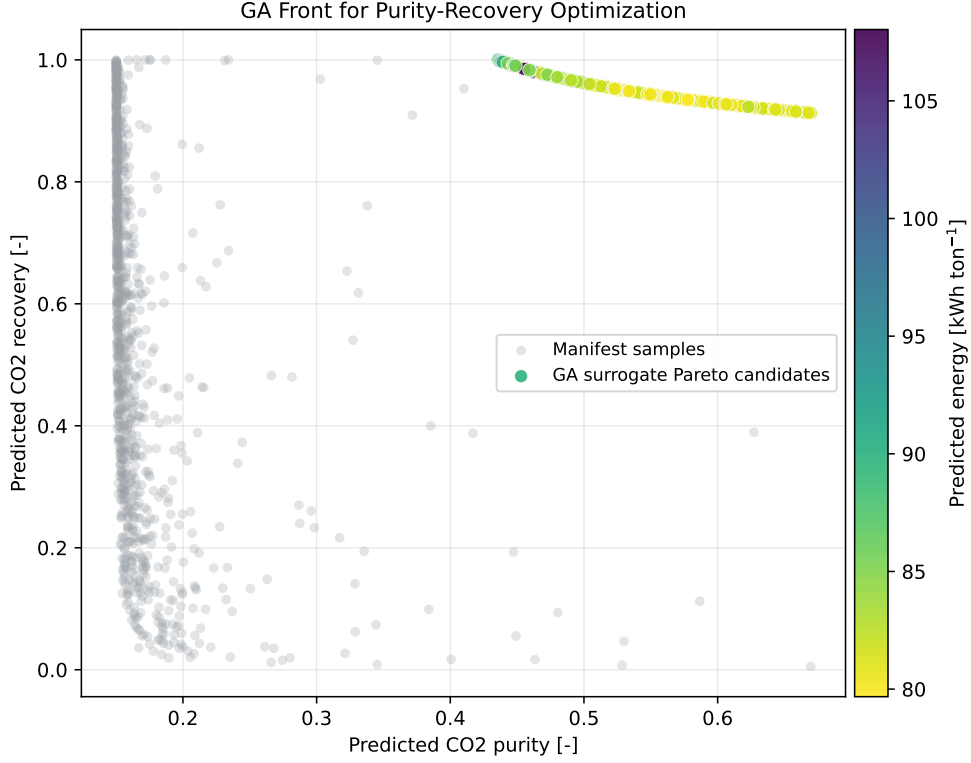


Figure 4: NSGA-II surrogate Pareto candidates for the purity-recovery problem.

For the productivity-energy problem, the Pareto set spans productivity from 16.49 to 115.45 mol kg⁻¹ h⁻¹ and energy from 36.70 to 266.08 kWh ton⁻¹. These candidates stay close to feed-level purity, so this front is mainly useful for high-throughput, low-energy screening rather than CO₂ enrichment.

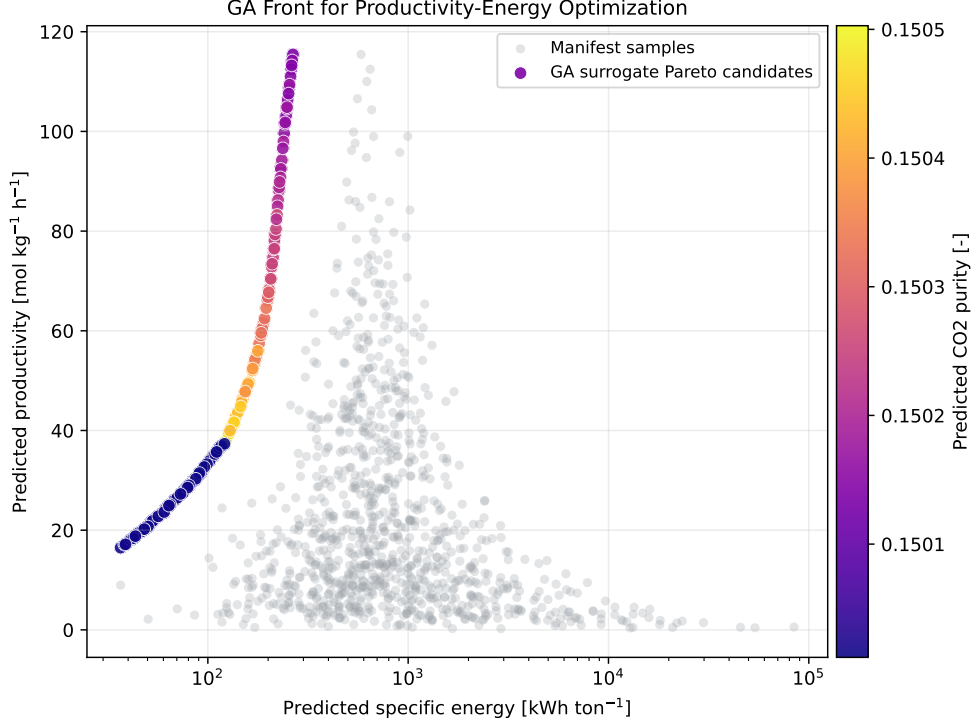


Figure 5: NSGA-II surrogate Pareto candidates for the productivity-energy problem.

6 Closed-Loop Detailed-Model Interface

The closed-loop handoff is file based. The GA writes:

- `data/ga_optimization_all.csv`: final GA populations for both problems.
- `data/ga_optimization_nondominated.csv`: final nondominated candidates.
- `data/detailed_model_input.csv`: 40 representative candidates for detailed PSA reruns.
- `data/detailed_model_results_template.csv`: the expected detailed-model result format.

After detailed simulations are completed, their results should be saved as `data/detailed_model_results.csv`. The comparison script will then generate `data/detailed_model_comparison.csv` and error summaries.

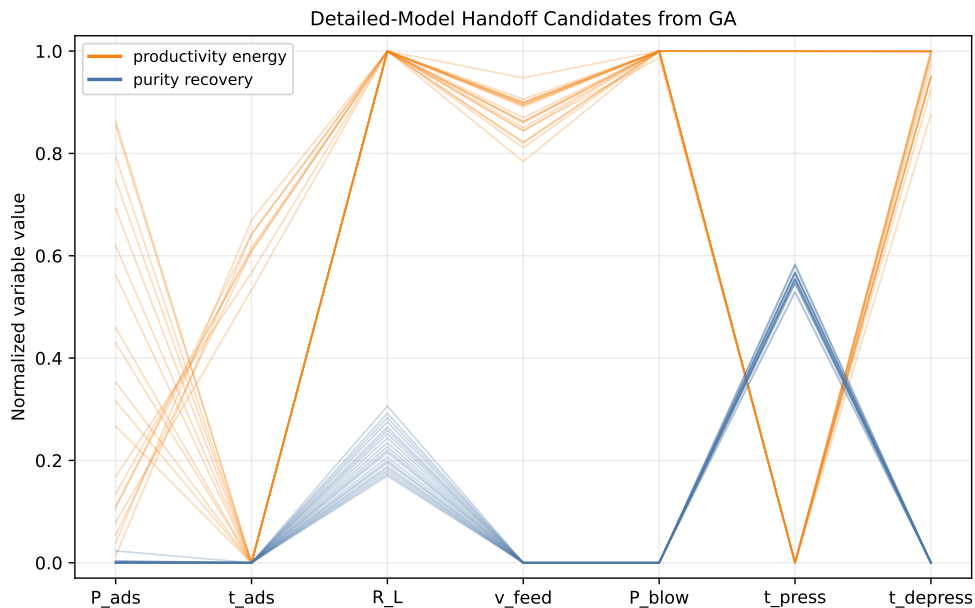


Figure 6: Current closed-loop handoff candidates. This figure is replaced by prediction-versus-detailed comparisons when detailed results are available.

7 Conclusions

The project now follows a closed, manifest-only route: surrogate training, GA-based multi-objective optimization, candidate export, and detailed-model comparison. The surrogate is strongest for recovery, productivity, and log energy. Purity is harder to learn from the scalar manifest alone, so optimized purity candidates should be treated as screening points that require detailed-model confirmation. The two GA fronts reveal different operating regimes: purity-recovery optimization favors low-throughput enriched cases, while productivity-energy optimization favors feed-level purity with much higher throughput.

AI Use Acknowledgement

Generative AI was used to assist with code organization, figure generation, and report drafting. The data, interpretation, and final responsibility remain with the student.